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COURSE REPORT
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# Antimicrobial Resistance Prediction from Bacterial Genome

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# Abstract

This report presents a determinant-based workflow for predicting antimicrobial resistance in *Salmonella enterica* from bacterial genomic data. The main objective is to determine whether curated resistance determinants, including acquired resistance genes and resistance-associated mutations, can be used to predict binary resistant/susceptible phenotypes for individual antibiotics. Instead of relying on genome-wide feature spaces that are harder to interpret biologically, the study focuses on a curated determinant space derived from AMRFinderPlus and MicroBIGG-E.

The workflow has three main parts: cleaning phenotype labels from AST data, cleaning and normalizing genotype records, and building one model-ready table for each antibiotic. To study how much biological knowledge should be used before learning starts, the report compares three connection scopes: *strict*, *broad*, and *all*. XGBoost models are then trained for each antibiotic with five-fold cross-validation. In addition, two simple rule-based baselines, *strict rule-based* and *broad rule-based*, are used as direct references without machine learning.

The results show that *all ML* gives the best overall predictive performance, while *broad ML* gives the best balance between feature coverage and biological interpretability. The rule-based baselines also show that, for some antibiotics, known genes and mutations are already enough for very accurate prediction. For other antibiotics, however, this direct gene-to-label rule is too simple and produces many false positives or false negatives. In those cases, machine learning helps by combining several signals at the same time.

Overall, the study shows that AMR prediction from curated genes and mutations is a suitable bioinformatics approach for this problem. It also shows that the difference between *strict*, *broad*, and *all* is important: these settings represent different trade-offs between biological simplicity, ease of interpretation, and predictive accuracy.

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# Glossary

| Term | Description |
|---|---|
| AMR | Antimicrobial resistance. |
| AST | Antimicrobial susceptibility testing. |
| Genotype | DNA-based representation of an isolate, here summarized by detected resistance genes and mutations. |
| Phenotype | Observed resistant or susceptible label from laboratory testing. |
| Determinant | A curated resistance gene or resistance-associated mutation. |
| MIC | Minimum inhibitory concentration, the drug level needed to stop visible bacterial growth. |
| BioSample | The isolate identifier used to match genotype and phenotype records. |
| Beta-lactam | A major antibiotic class that includes drugs such as penicillins and cephalosporins. |
| Strict scope | Uses only determinants closely matched to the target antibiotic. |
| Broad scope | Uses determinants from the broader drug class of the target antibiotic. |
| All scope | Uses all curated AMR determinants detected in the isolate. |
| ROC-AUC | A score that measures how well a model separates resistant and susceptible isolates across thresholds. |
| Zero-feature row | A sample whose input vector contains no matched determinant evidence under a given scope. |

# Chapter 1

# Introduction

## 1.1 Problem Definition

Antimicrobial resistance (AMR) prediction from genomic data is a problem of predicting a laboratory outcome from DNA-based evidence. In this report, the target organism is *Salmonella enterica*, an important foodborne bacterial pathogen that is often monitored in AMR surveillance programs. The practical objective is simple: given the genomic content of one isolate, can we predict whether it will be resistant or susceptible to a specific antibiotic in laboratory testing?

The input side of the problem is the genotype representation of each isolate. Instead of using the raw genome sequence directly, this study uses a curated list of detected AMR genes and resistance-related mutations from AMRFinderPlus and the NCBI MicroBIGG-E resource. Each record also includes useful labels such as drug class, drug subclass, detection method, and evidence scope. This makes the features easier to understand than genome-wide features such as generic k-mers, which are only short sequence fragments without direct biological meaning on their own.

The output side of the problem is the phenotype label obtained from antimicrobial susceptibility testing (AST). For each antibiotic, the final target is binary: resistant isolates are positive cases and susceptible isolates are negative cases. Intermediate and ambiguous labels are removed so that the prediction target stays clear and easy to interpret.

The central research question of the project is therefore:

> Can curated AMR determinants from a *Salmonella enterica* genome reliably predict antibiotic resistance phenotype, and how does prediction quality change under different levels of biological constraint?

This question matters from both the biological and computational sides. Biologically, known resistance genes and mutations are important, but gene presence alone does not always explain the final laboratory result. Computationally, the project must decide how strictly it should follow known antibiotic groupings before machine learning is applied.

The work in this report studies that design space through three feature-connection settings:

- **Strict**: only determinants directly aligned with the narrow lineage of the target antibiotic are used.
- **Broad**: determinants from the broader drug class of the target antibiotic are included.
- **All**: all curated AMR determinants observed in the isolate are included, regardless of narrow lineage.

These settings allow the report to study not only prediction quality, but also the trade-off between simple biological interpretation and broader feature coverage.

## 1.2 Related Work

Research on genomic AMR prediction can be grouped into three broad method families.

### 1.2.1 Biology-first and rule-based studies

The first family starts from known AMR genes and known resistance mutations, then predicts phenotype directly from curated biological knowledge. The main advantage is simplicity: every prediction can be traced back to a known gene or mutation. In addition, tools such as AMRFinderPlus provide a curated catalog of resistance mechanisms and standardized annotations that are useful for surveillance and genotype screening.

However, rule-based approaches are limited by what is already known. If resistance depends on unknown variants, background lineage effects, interactions between genes, or incomplete annotation, a purely rule-based system may miss resistant isolates or may not explain mismatches well. For the present project, this family motivates the use of a curated determinant-based rule baseline rather than a manually designed expert system.

Representative studies and resources in this direction include the AMRFinderPlus framework of Feldgarden et al. [1] and the genotype-phenotype comparison study of Shen et al. [2].

### 1.2.2 Biology-informed machine learning

The second family uses biologically meaningful features but lets machine learning learn the mapping from genotype to phenotype. This direction is especially relevant to the current study. Instead of relying only on direct rules, a model is trained on AMR genes, mutations, and related annotations that already have biological meaning.

The strength of this family is balance. Feature names such as resistance genes or point mutations remain meaningful, while the model can still learn combinations and patterns that are hard to encode as fixed rules. Studies in this family are therefore close in spirit to the present work, especially when the goal is to keep mechanism-level interpretation instead of treating the task as a pure black-box prediction problem.

Examples include the work of Maguire et al. [3], who analyzed major resistance drivers in nontyphoidal *Salmonella*, and the more recent Salmonella-focused machine learning studies of Benefo et al. [4] and He et al. [5].

### 1.2.3 Genome-wide and ML-heavy approaches

The third family uses broad genome-wide representations such as SNPs, k-mers, pan-genome features, or graph-based encodings. These methods may capture wider genomic signal than curated determinant-only pipelines and can help when resistance depends on broader sequence context.

Their main limitation is reduced biological transparency. A genome-wide feature with strong predictive value does not always correspond to a direct resistance mechanism. Such models may learn lineage background or dataset-specific correlations rather than a clear AMR mechanism. For this reason, they are valuable as high-coverage predictive approaches, but they are less convenient when the report needs to explain why a prediction was made in biological terms.

Representative examples of this broader direction include the whole-genome machine learning framework of Ren et al. [6] and the k-mer-based MIC prediction study of Wang et al. [7].

### 1.2.4 Position of the present study

This report belongs primarily to the biology-informed machine learning family. It uses curated AMR determinants from AMRFinderPlus and MicroBIGG-E as interpretable features, while also comparing different levels of biological constraint in the genotype-to-phenotype connection step. In addition, the report includes a curated determinant-based rule baseline to show how far direct genotype-to-phenotype inference can go before machine learning is introduced.

The study therefore occupies an intermediate position between strict rule-based inference and broad genome-wide prediction. This position is appropriate for a bioinformatics report because it emphasizes both predictive performance and biological explanation.

## 1.3 Research Objectives

The report is guided by the following objectives:

1. Build a reproducible pipeline that connects assembled bacterial genomes, curated AMR determinant detection, AST phenotype labels, and antibiotic-specific prediction datasets.

2. Clean and normalize genotype and phenotype data so that the final cohort contains only isolates with usable and interpretable information.

3. Construct three determinant-connection settings, namely *strict*, *broad*, and *all*, in order to study how biological scope affects prediction quality.

4. Train and evaluate antibiotic-specific machine learning models on the prepared datasets using a consistent cross-validation protocol.

5. Add a curated determinant-based rule baseline to represent direct genotype-to-phenotype inference without machine learning.

6. Compare rule-based and machine-learning approaches in terms of recall, precision, and overall predictive performance.

7. Explain the biological meaning of the observed differences, especially the trade-off between interpretability, feature coverage, and genotype-phenotype mismatch.

Taken together, these objectives define two concrete deliverables. First, the study provides a methodology that can be reproduced and extended. Second, it provides a clear comparison showing whether broader determinant usage and machine learning offer benefits beyond direct rule-based inference.

# Chapter 2

# Biological Background and Data Description

## 2.1 Biological background of antimicrobial resistance

Antimicrobial resistance emerges when bacteria acquire genetic changes that reduce or eliminate the effect of antibiotic treatment. These changes may come from acquired resistance genes, point mutations in native genes, gene amplification, or broader genomic context affecting gene regulation. In *Salmonella enterica*, resistance can involve beta-lactamases, aminoglycoside-modifying enzymes, quinolone-related point mutations, tetracycline resistance genes, and many other mechanisms. Antimicrobial resistance appears when bacteria gain genetic changes that weaken or block the effect of an antibiotic. In practice, these changes are often known resistance genes or specific mutations. In *Salmonella enterica*, resistance can come from several well-known mechanisms, for example enzymes that break down beta-lactam drugs such as penicillins and cephalosporins, genes that protect against tetracycline, or mutations linked to quinolone resistance.

For this reason, AMR prediction from genomic data is not only a classification task. It is also an interpretation task. A useful AMR predictor should not only output a label, but should also keep a clear link between that label and known resistance genes or mutations.

## 2.2 Phenotype definition: AST and resistance labels

The phenotype side of the project comes from antimicrobial susceptibility testing. AST checks whether an isolate behaves as resistant or susceptible to a specific drug in laboratory conditions. The raw measurement is often the minimum inhibitory concentration (MIC), which is the drug level needed to stop visible bacterial growth. This value is then converted into a label such as susceptible, intermediate, or resistant according to a breakpoint standard.

In this project, the final supervised learning label is binary:

- **Resistant** is encoded as the positive class.
- **Susceptible** is encoded as the negative class.

Intermediate, nonstandard, and ambiguous labels are excluded to avoid adding noise near the decision boundary. This choice fits the current report because the goal is a robust binary prediction task rather than detailed MIC prediction.

## 2.3 Genotype definition: curated AMR determinants

The genotype side of the project is built from curated AMR determinant resources rather than raw sequence fragments. Each isolate is represented by detected resistance genes and mutations reported by AMRFinderPlus and organized in the MicroBIGG-E resource. This is useful because AMRFinderPlus provides a curated catalog and structured annotations for resistance determinants [1]. Each determinant record contains several important fields:

- `Element symbol`: the name of the gene or mutation.
- `Class`: a broad drug class.
- `Subclass`: a more specific drug group.
- `Subtype`: whether the determinant is a gene or a mutation.
- `Method`: how the determinant was detected.
- `Scope`: how conservative the evidence is.

This representation fits the report well because every feature still has a biological meaning and can be traced back to a known resistance mechanism.

## 2.4 Data sources used in the project

The project uses three major data sources:

1. **Assembled genomes in FASTA format**, used for AMRFinderPlus-based prototype validation.
2. **NCBI MicroBIGG-E genotype data**, used as the main large-scale determinant table for feature construction.
3. **NCBI AST phenotype data**, used as the ground-truth source of antibiotic resistance labels.

The common key connecting these resources is the `BioSample` identifier. It allows genotype and phenotype rows to be matched isolate by isolate and makes antibiotic-specific datasets possible.

## 2.5 Data filtering and cohort selection

Not every downloaded record is suitable for training. The report therefore relies on a filtered and aligned cohort with the following principles:

- keep only isolates that appear in both genotype and phenotype resources;
- keep only antibiotics with enough observations for stable model training;
- keep only phenotype rows with usable binary resistance labels;
- keep only AMR-related genotype rows with biologically meaningful evidence.

For the final modeling cohort, antibiotics must satisfy minimum sample thresholds so that each target has enough resistant and susceptible examples. This is necessary because resistance rates differ strongly between drugs, and some drugs simply do not have enough data for stable model evaluation.

## 2.6 Why this dataset is biologically non-trivial

Even with curated determinant features and cleaned phenotype labels, genotype does not map perfectly to phenotype. An isolate may carry a resistance gene but still test susceptible, or it may test resistant even when no obvious determinant is detected in the curated feature set. Several biological reasons can explain this mismatch:

- a gene may be present but not strongly expressed;
- different variants of the same gene may behave differently;
- multiple genes may interact with each other;
- the curated catalog may still miss some resistance signals;
- laboratory measurement and breakpoint rules may also affect the final label.

This genotype-phenotype gap is one of the main reasons why both rule-based inference and machine learning must be tested carefully instead of being assumed to work perfectly.

# Chapter 3

# Methodology

## 3.1 Model Training and Prediction

The main predictive model in this project is an antibiotic-specific XGBoost classifier trained on prepared feature tables built from detected genes and mutations. One model is trained for each antibiotic because the relationship between genotype and phenotype can differ greatly from one drug family to another.

The project uses a supervised binary classification setting with resistant isolates treated as the positive class. This choice matches the practical objective of detecting resistant cases. The feature matrix contains determinant-level evidence from the normalized genotype table, while the target labels come from cleaned AST phenotype data.

Cross-validation is used to estimate model performance consistently across antibiotics. The current implementation uses five-fold stratified cross-validation with a fixed random seed. This keeps the class ratio similar across folds and makes the evaluation more stable than a single train-test split. The main XGBoost settings are: `n_estimators = 250`, `learning_rate = 0.05`, `max_depth = 5`, `subsample = 0.8`, and `colsample_bytree = 0.8`. The implementation also uses class weighting through `scale_pos_weight` so that resistant cases are not dominated by the larger susceptible class when the data are imbalanced.

The evaluation focuses on the following metrics:

- **Recall**: important because missing a resistant isolate is a high-risk error.
- **Precision**: important because incorrectly calling an isolate resistant also has practical cost.
- **ROC-AUC**: useful for measuring how well the model separates resistant and susceptible isolates across thresholds.
- **Accuracy**: reported as supporting context rather than the main metric.

In addition to machine learning, the final evaluation includes curated determinant-based rule baselines. These baselines use the same cleaned cohort and the same prepared input tables as the ML models, but replace learned prediction with a direct rule. For one isolate under a given scope, the prediction is resistant if at least one connected gene- or mutation-based feature is present; otherwise the prediction is susceptible. Under this design, the difference between *strict rule-based* and *broad rule-based* comes from the feature scope in the prepared input tables rather than from a different decision formula.

This setup is important because it separates two sources of improvement:

- improvement gained by relaxing or tightening the biological scope; and
- improvement gained by replacing a direct rule with machine learning.

As a result, the report can compare a simple biology-based rule against a learned model under the same dataset construction protocol.

## 3.2 Data Processing and Pipeline Construction

The full methodology of the project is not a single training script but a staged bioinformatics pipeline. The workflow begins with assembled genomes and curated NCBI resources, continues through genotype and phenotype cleaning, and ends in antibiotic-specific prediction tables suitable for machine learning and rule-based evaluation. The purpose of this chapter is to describe that transformation in a reproducible order.

### 3.2.1 Solution Integration

The proposed solution combines biological data sources, curated AMR annotations, and predictive modeling into one workflow. Instead of training directly on raw sequence, the pipeline first builds an interpretable feature space based on detected resistance genes and mutations, and then connects that feature space to antibiotic-specific phenotype labels.

At a high level, the integrated pipeline can be summarized as follows:

1. obtain determinant data from AMRFinderPlus and MicroBIGG-E;
2. obtain phenotype labels from AST records;
3. align both resources by `BioSample`;
4. clean and normalize genotype and phenotype tables;
5. construct antibiotic-specific feature tables under different biological scopes;
6. train and evaluate prediction models;
7. compare model-based inference with curated determinant-based rule inference.

This integration is important because every stage remains traceable. The final prediction is not separated from its biological origin. The report can still explain where each feature comes from, how each label is defined, and why a particular set of genes or mutations is considered relevant to a target antibiotic.

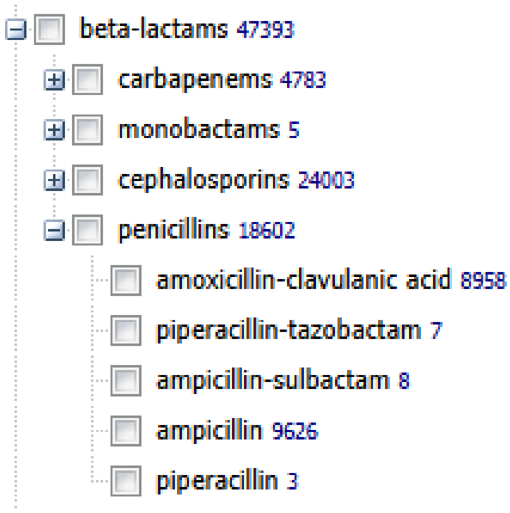
### 3.2.2 Workflow Construction

The workflow is constructed in four major stages.

**Stage 1: phenotype cleaning.** Phenotype processing is performed first because the task is defined by antibiotic-specific labels. Rows with missing key fields, ambiguous labels, or contradictory duplicates are removed. Only resistant and susceptible labels are kept for binary modeling. Antibiotics with too few records or too few resistant examples are excluded from the final cohort.

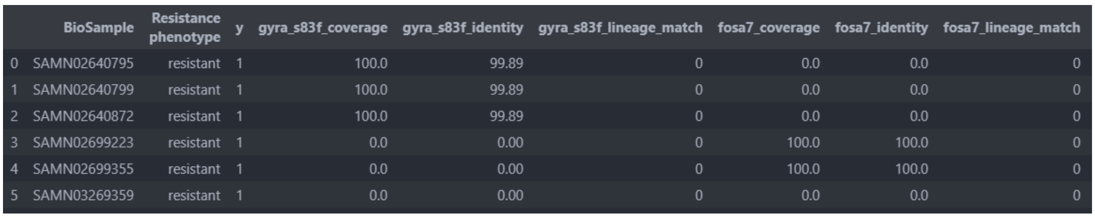
**Stage 2: genotype cleaning.** Genotype processing starts from determinant-level records. Exact duplicates are removed. If the same gene or mutation is detected multiple times in one isolate, only one preferred record is kept. Noisy categories are filtered out, and class and subclass labels are normalized so that later matching against antibiotic groups stays consistent.

**Stage 3: genotype-to-phenotype connection.** For each antibiotic, genotype rows are connected to phenotype labels under one of three scopes: *strict*, *broad*, or *all*. This is the key design choice of the project because it controls how narrowly or broadly the features are selected before learning begins.




Hình 3.1: Broad class hierarchy for beta-lactam antibiotics.

**Stage 4: model-ready table generation.** After connection, the genotype evidence is converted into antibiotic-specific feature tables. These tables are the final inputs for XGBoost and for the rule-based baselines. Each table represents one modeling problem for one antibiotic under one biological scope, with one row per isolate.



| | BioSample | Resistance phenotype | y | gyra_s83f_coverage | gyra_s83f_identity | gyra_s83f_lineage_match | fosa7_coverage | fosa7_identity | fosa7_lineage_match |
|---|---|---|---|---|---|---|---|---|---|
| 0 | SAMN02640795 | resistant | 1 | 100.0 | 99.89 | 0 | 0.0 | 0.0 | 0 |
| 1 | SAMN02640799 | resistant | 1 | 100.0 | 99.89 | 0 | 0.0 | 0.0 | 0 |
| 2 | SAMN02640872 | resistant | 1 | 100.0 | 99.89 | 0 | 0.0 | 0.0 | 0 |
| 3 | SAMN02699223 | resistant | 1 | 0.0 | 0.00 | 0 | 100.0 | 100.0 | 0 |
| 4 | SAMN02699355 | resistant | 1 | 0.0 | 0.00 | 0 | 100.0 | 100.0 | 0 |
| 5 | SAMN03269359 | resistant | 1 | 0.0 | 0.00 | 0 | 0.0 | 0.0 | 0 |

Hình 3.2: Example of a prepared model-input table after genotype evidence has been aligned with phenotype labels.

This staged construction lets the report separate data cleaning decisions from modeling decisions and explain each step clearly.

### 3.2.3 Workflow Evolution

The workflow did not begin in its final form. It evolved from a simple determinant extraction idea into a structured comparison framework. This evolution matters because it explains why the final report compares multiple biological scopes instead of presenting only one model.

The earliest idea was that only determinants directly tied to a target antibiotic should be used. This led naturally to the *strict* setting. However, exploratory analysis showed that this setting often leaves many isolates with no usable determinant evidence, even when those isolates belong to the valid phenotype cohort. In practical terms, the resulting feature tables contain many all-zero rows, which makes the narrow design too fragile for several antibiotics.

To address this issue, a broader within-class setting was introduced. The *broad* scope allows genotype evidence from the larger drug class rather than only the narrow lineage branch. This improves coverage while still keeping the feature space guided by biological knowledge.

Finally, the *all* setting was added as the widest determinant-based representation. This setting does not restrict the features to one drug lineage and therefore tests whether the model benefits from the full curated set of detected resistance determinants in each isolate. The purpose of *all* is not to maximize interpretability, but to measure how much predictive benefit appears when biological constraints are relaxed.

This evolution from *strict* to *broad* to *all* forms the core experimental design of the project. It should be understood as a deliberate trade-off between simplicity, coverage, and predictive power rather than as a purely technical parameter sweep.

### 3.2.4 Workflow Execution

In execution, the pipeline produces one prepared dataset per antibiotic per biological scope. Each prepared table contains one row per isolate and a set of feature columns derived from detected genes and mutations. The current implementation uses coverage, identity, and lineage-related evidence from the curated determinant table as model inputs.

Workflow execution can be understood at two complementary levels:

- **Data preparation level**: how the phenotype and genotype tables are transformed into prepared model-input tables.
- **Modeling level**: how prepared tables are passed into antibiotic-specific training and evaluation.

The codebase also separates data ingestion, staged storage, prepared-table construction, and prediction logic. Although the main focus of this report is scientific evaluation rather than software architecture, this staged design supports reproducibility and keeps biological preprocessing separate from prediction-serving interfaces.

By the end of this workflow, the cohort, feature scopes, evaluation protocol, and comparable outputs are all defined explicitly, which makes the later experimental comparison straightforward.

# Chapter 4

# Experiments and Results

## 4.1 Experimental Results

### 4.1.1 Experimental setup

The experiments are conducted on antibiotic-specific datasets prepared from the same shared genotype-phenotype cohort. The same cohort selection rules, phenotype cleaning rules, and genotype normalization rules are applied across all scopes to keep the comparison fair. The current implementation evaluates 14 antibiotics that satisfy the minimum sample and class-count thresholds required for stable supervised learning.

The main model-based comparison includes three determinant-connection scopes:

- **Strict ML**
- **Broad ML**
- **All ML**

The direct genotype-to-phenotype baseline is implemented in two forms:

- **Strict rule-based**
- **Broad rule-based**

This comparison is more meaningful for this bioinformatics study than comparing several generic classifiers because it isolates the effect of biological scope and prior knowledge.

### 4.1.2 Average performance across methods

Table 4.1 summarizes the average performance of the five main method settings across the 14 antibiotics.

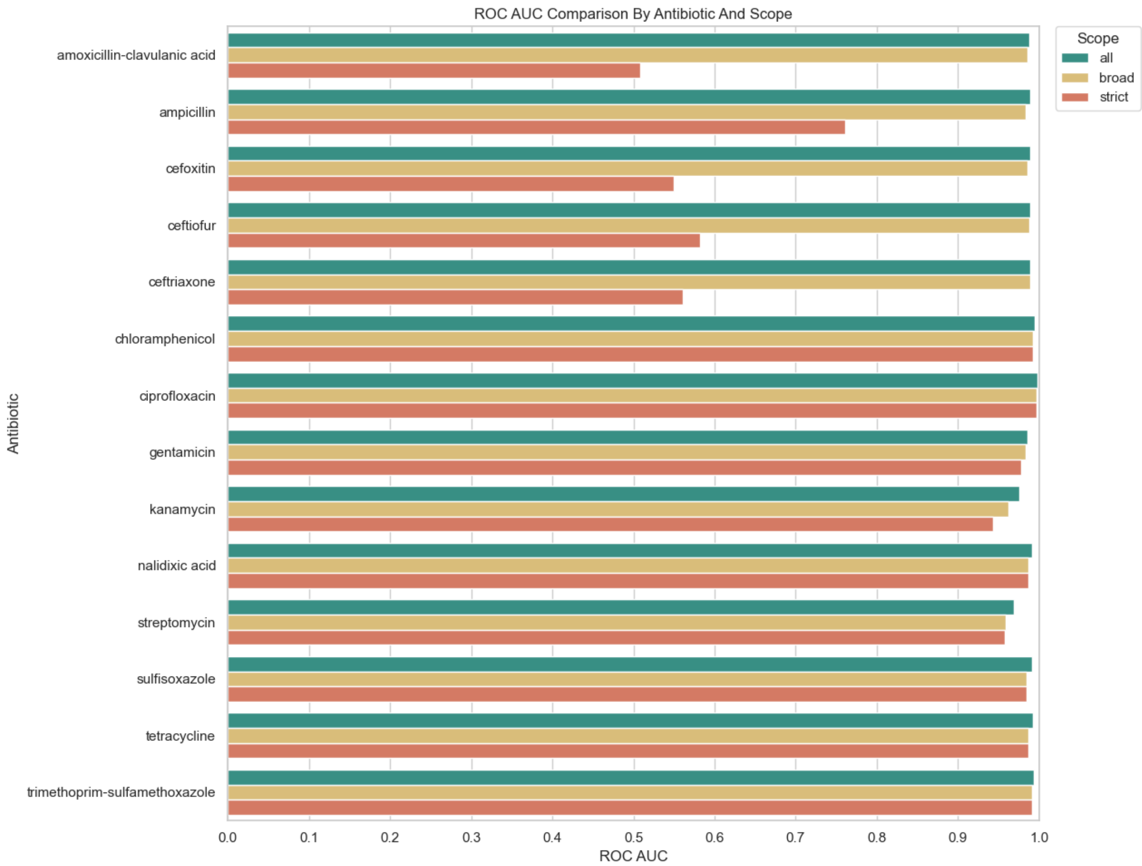
Bảng 4.1: Average performance across antibiotics

| Method | Accuracy | Precision | Recall | ROC-AUC | Avg. zero rows |
|---|---|---|---|---|---|
| Strict rule-based | 0.861 | 0.655 | 0.704 | – | 4461.1 |
| Broad rule-based | 0.839 | 0.648 | 0.987 | – | 3790.5 |
| Strict ML | 0.822 | 0.744 | 0.865 | 0.842 | 4461.1 |
| Broad ML | 0.988 | 0.972 | 0.974 | 0.984 | 3790.5 |
| All ML | 0.987 | 0.966 | 0.974 | 0.988 | 0.0 |

Several patterns are immediately visible. First, *broad ML* and *all ML* are the strongest overall settings, both achieving average accuracy near 0.99 and average recall around 0.974. Second, *broad rule-based* has the highest average recall among the direct rule methods, reaching 0.987, but this comes with a clear loss in precision and specificity. Third, the *strict* settings use a narrower biological scope but suffer from many zero-hit or zero-feature rows, especially for some beta-lactam antibiotics.

### 4.1.3 Observed performance pattern of the ML scopes

The model-based results show a clear overall pattern. The *all* scope gives the best ability to separate resistant and susceptible isolates across antibiotics and removes the zero-feature problem completely. Its average ROC-AUC is 0.988, higher than the 0.984 achieved by *broad ML* and much higher than the 0.842 of *strict ML*. The *broad* scope performs competitively for many antibiotics and is the strongest constrained alternative. The *strict* scope is still useful as a narrow reference, but it is too fragile to serve as the default setting because it leaves too many isolates with no usable determinant evidence.




Hình 4.1: ROC-AUC comparison across antibiotics and determinant-connection scopes.

This pattern is biologically interpretable. Narrow matching improves feature specificity, but it can also exclude genes or mutations that are still useful within the same broader drug family. Relaxing the scope to *broad* captures more same-class mechanisms, while the *all* setting allows the model to use wider patterns among many detected determinants.

### 4.1.4 Main result trends

The results support four main conclusions:

1. The *all* scope is the strongest global default when the main objective is predictive performance.
2. The *broad* scope is the best compromise when interpretability is still important.

3. The *strict* scope is useful as a narrow biological baseline, but it fails on several antibiotics because the matching rule is too restrictive.

4. Performance differences across antibiotics reflect real differences in resistance mechanism diversity and determinant coverage.

### 4.1.5 Rule-based versus machine-learning behavior

The rule-based baselines are useful because they show when the presence of known determinants is already enough to predict phenotype and when machine learning adds real value. For some antibiotics, the direct rule is already very strong. For example, under the strict scope, ciprofloxacin reaches precision 0.968 and recall 0.998, chloramphenicol reaches precision 0.990 and recall 0.988, and nalidixic acid reaches precision 0.953 and recall 0.989. These results suggest that, for such antibiotics, the detected genes and mutations already match the phenotype closely.

However, the same direct rule becomes too weak or too coarse for other antibiotics. Under the strict rule baseline, ceftriaxone reaches recall only 0.101 and precision 0.090, while ceftiofur reaches recall 0.133 and precision 0.102. When the scope is relaxed to broad rule-based inference, recall for ceftriaxone increases sharply to 0.985, but precision remains only 0.489. This shows the main trade-off clearly: broader matching can recover missed resistant isolates, but it can also overcall resistance by accepting determinants that are related to the same drug class but are not specific enough to the target drug.

Machine learning changes this balance substantially. For ceftriaxone, *broad ML* achieves precision 0.992 and recall 0.976, while *all ML* achieves precision 0.990 and recall 0.974. In other words, the learned models keep the broad coverage needed to recover resistant isolates, but avoid much of the precision loss seen in broad rule-based inference. A similar pattern appears for ceftiofur and cefoxitin. These comparisons support the main scientific argument of the report: machine learning is most valuable when broader determinant evidence must be combined more carefully than a direct rule can do.

### 4.1.6 Interpretation of zero-hit and zero-feature rows

One of the clearest signals in the experiment is the role of zero rows. In the rule-based setting, a zero-hit row means that no determinant connected to the target antibiotic scope was found in the isolate. In the machine-learning setting, the corresponding zero-feature row means the model received an all-zero feature vector for that isolate. The averages in Table 4.1 show that both strict settings suffer from roughly 4461 zero rows on average, whereas broad settings reduce this burden to about 3790 rows, and the all scope removes it entirely.

This result explains much of the performance gap between strict and broader settings. A narrow scope can be biologically conservative, but if it leaves too many isolates without usable determinant evidence, both rule-based prediction and machine learning start from a weak input.

### 4.1.7 Discussion of limitations

The current study predicts a binary resistant/susceptible phenotype rather than MIC, uses curated determinant resources rather than full pan-genome features, and has not yet performed external validation on an independent surveillance cohort. These limitations do not invalidate the study, but they define its scope clearly and position it as a determinant-based bioinformatics analysis rather than a full clinical deployment study.

# Conclusion and Future Work

This report studies antimicrobial resistance prediction for *Salmonella enterica* using curated AMR determinants derived from genomic data. The work combines biological filtering, determinant normalization, genotype-phenotype cohort alignment, and antibiotic-specific predictive modeling into a reproducible pipeline. Rather than treating the task as a generic machine learning problem, the report studies how biological scope affects prediction quality.

The main methodological contribution is the comparison of three determinant-connection scopes: *strict*, *broad*, and *all*. The current results indicate that the *all* scope provides the strongest global predictive performance, while the *broad* scope offers the most practical compromise between biological interpretability and model effectiveness. The *strict* scope remains useful as a narrow biological reference, but it is too restrictive to serve as a universal default across antibiotics.

Another important conclusion is that feature-engineering decisions are not merely technical. Choices such as phenotype cleaning, determinant filtering, class/subclass normalization, and antibiotic-scope matching directly shape the biological meaning of the resulting model inputs. For that reason, the pipeline is best understood as a bioinformatics workflow rather than only as a classifier benchmark.

The addition of curated determinant-based rule baselines under the *strict* and *broad* settings further clarifies the main result. Direct genotype-to-phenotype inference is already highly effective for some antibiotics, but it becomes too coarse when broader class-level matching is required. In those settings, machine learning keeps broad biological coverage while sharply improving precision. This comparison shows that the main value of learned models lies not in replacing biological knowledge, but in combining broader determinant evidence more effectively than a direct rule can.

Future work may include MIC prediction, external validation on an independent cohort, more detailed interpretation of top-ranked determinants, and broader resistome or pan-genome feature integration.

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